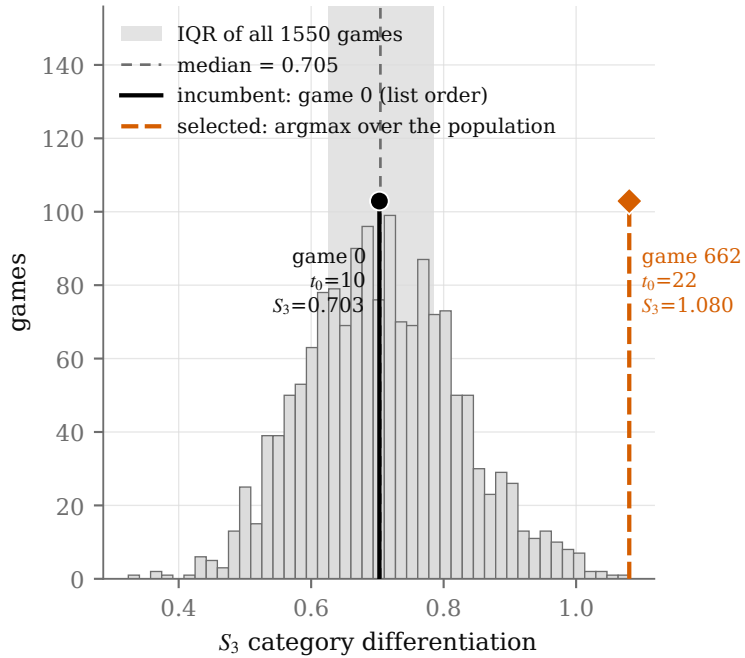
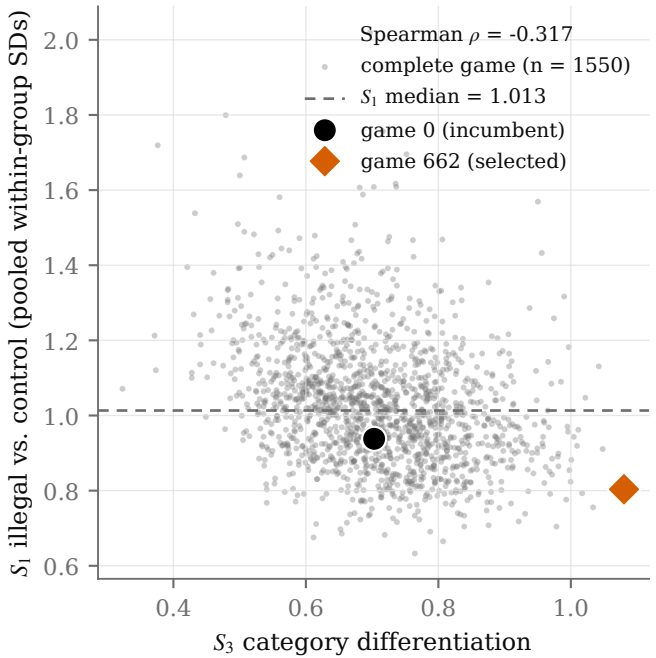
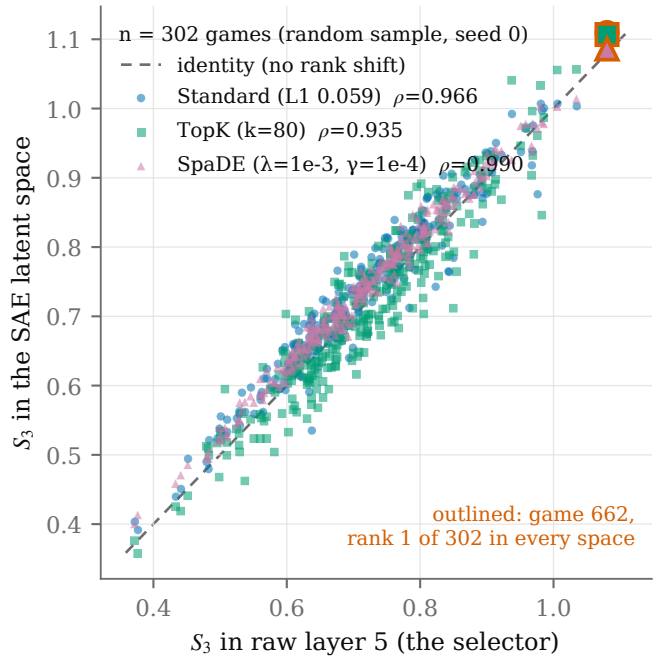






S_3 — mean pairwise $\|X_i - X_j\|$ over the six illegal branches ÷ RMS $\|X\|$; 0 = they displace identically, $\sqrt{2}$ = mutually orthogonal.

S_1 — $\|\mu_{\text{illegal}} - \mu_{\text{control}}\|$ ÷ pooled within-group RMS radius, same stacked vectors.

(a) and (b) are the whole population of complete games, not a sample: the spread is the population IQR, not an error bar.